

LAB1: Software Engineering

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Section: D

Webhook Ingestion & Retry Mechanism Hub

Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest inbound webhooks, validate the request payload, and record the webhook event for processing.	High	Pass: Valid webhook is accepted and logged with its payload. Fail: Invalid webhook is processed as a valid event.	Ensures webhook events are received and recorded correctly.
FR-002	Functional	The system shall forward valid webhook payloads to the configured target internal services.	High	Pass: Valid payload reaches the configured target endpoint. Fail: Payload is not delivered to the target service.	Allows third-party events to reach the required internal services.
FR-003	Functional	The system shall automatically retry failed webhook deliveries using exponential backoff intervals.	High	Pass: Failed deliveries are retried at increasing intervals and each attempt is logged. Fail: Failed delivery is dropped without retry.	Improves delivery reliability during temporary service failures.
FR-004	Functional	The system shall maintain logs containing webhook payload details, delivery attempts, responses, and processing status.	Medium	Pass: Required event and delivery information can be viewed in the logs. Fail: Important delivery information is missing.	Helps operators monitor, troubleshoot, and audit webhook processing.
FR-005	Functional	The system shall place repeatedly failed webhook events into a dead-letter queue for later review and controlled reprocessing.	Medium	Pass: Events exceeding the retry limit are stored in the dead-letter queue. Fail: Failed events are lost or cannot be reviewed.	Prevents permanent loss of events that cannot be delivered automatically.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The ingestion endpoint shall buffer inbound webhooks with 99.999% reliability and support throughput up to 2,500 requests per second.	High	Pass: Benchmarking confirms the target reliability and throughput under simulated peak load. Fail: The target throughput or reliability is not achieved.	Supports reliable ingestion during high-volume webhook traffic.
NFR-002	Reliability & Availability	The system shall preserve webhook	High	Pass: After a simulated service interruption,	Ensures resilience and prevents event loss

		events and retry state during service interruptions and recover processing without losing accepted events.		accepted events and retry state are recovered and processing resumes. Fail: Accepted events or retry information is lost.	during failures.
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Use-Case Flow Specification

Use Case: Process Webhook Event

Primary Actor	Integration Developer
Goal	To receive, process, deliver, and reliably retry an inbound webhook event.
Preconditions	1. The webhook endpoint is configured and available. 2. The target internal service is configured. 3. The incoming webhook contains the required request information.
Postconditions	1. The webhook event is delivered successfully or stored in the dead-letter queue after failed retries. 2. Delivery attempts and processing status are recorded in the logs. 3. The event is not lost after an accepted request.

Main Success Scenario

1. The system receives an inbound webhook from a third-party source.
2. The system validates the webhook request and records the incoming payload.
3. The system forwards the valid payload to the configured target internal service.
4. The target service responds successfully.
5. The system records the successful delivery attempt and processing status.
6. The webhook event is marked as successfully processed.

Alternate Flow

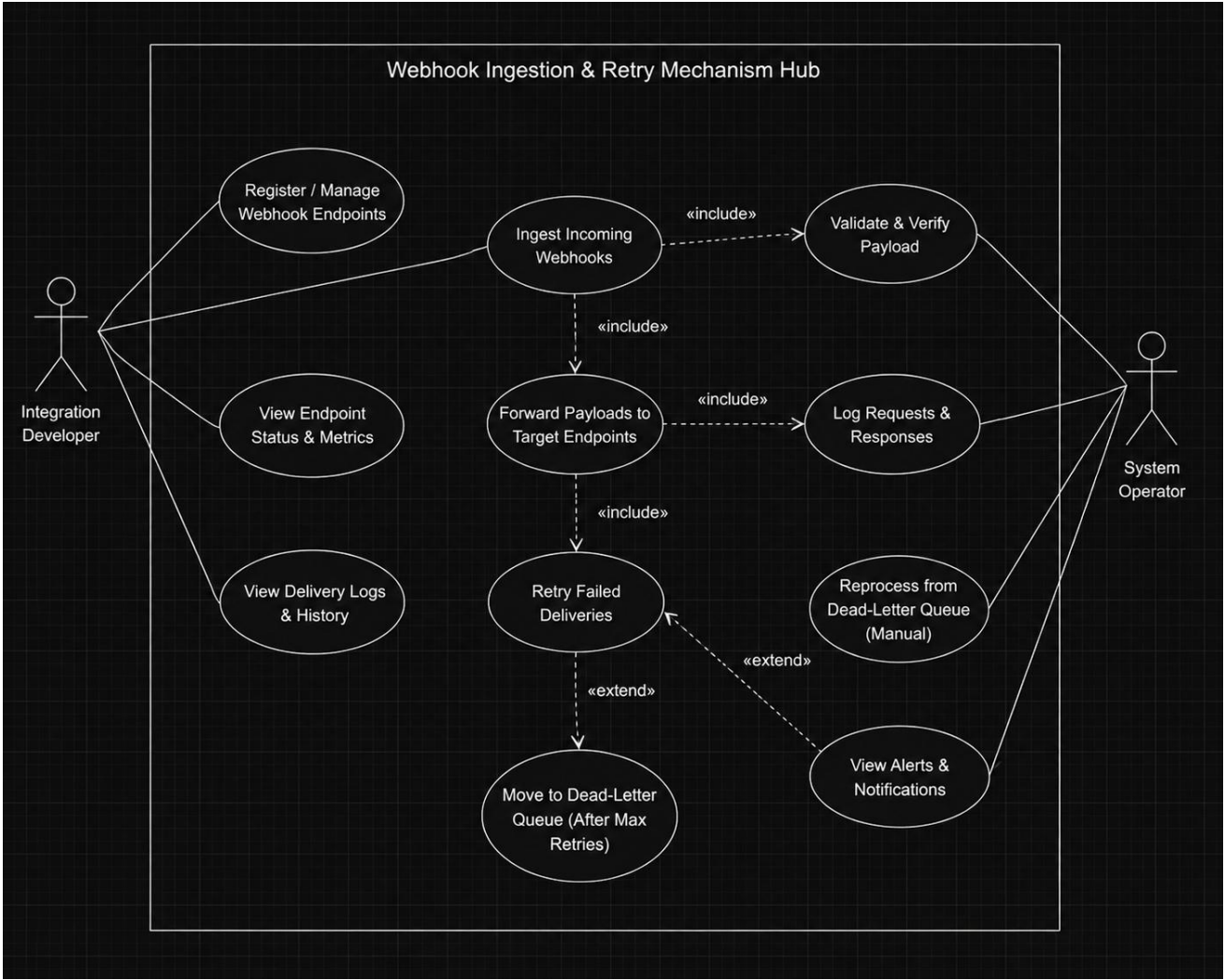
A1. Target Service Delivery Failure

1. The target service returns an error or does not respond.
2. The system records the failed delivery attempt.
3. The system retries delivery using exponential backoff.
4. If a later retry succeeds, the system records the successful delivery.
5. If the retry limit is reached, the system moves the event to the dead-letter queue.

A2. Invalid Webhook Payload

1. The system detects that the webhook payload is invalid.
2. The system rejects the request and records the validation failure.
3. The invalid event is not forwarded to the target service.

UML Use-Case Diagram



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